

7. GOVERNANCE & RELEASE GATE · STPA Hazard Mitigation ·
Bumpless Override · Defensible Release Case (**LOOP-01** → **REL-01**)

1. SENSING
Photons / Voltages to DMA
MIPI CSI-2 · SPI Bus Priority

2. PERCEPTION
ViT Encoders · DINOv2
3D Spatial Affordance Tokens

3. MEMORY
Temporal Belief · $SE(3)$ Trees
JEPA / RSSM World Models

5. PLANNING (System 1.5 · MPU/NPU)
Diffusion Policies · ACT Action Chunking
 H -Step Action Horizons · Jerk Continuity

4. REASONING (System 2 · MPU)
Vision-Language Foundation Models · 3D Goals
Leases with Expiring TTL (t_{expire})

THE PROPOSAL-PERMISSION PRIVILEGE BOUNDARY

6. REFLEX (System 1 · MCU)
1 kHz Real-Time Loop · Control Barrier Functions ($h(x) \geq 0$) · Stopping Distance $d_{\text{stop}} \leq d_{\text{clearance}} \cdot \text{Veto}(u_t)$

THE PHYSICAL WORLD ($W_t \rightarrow W_{t+1}$) · Kinetic Momentum · Joule Heat · Friction · Collision Dynamics

Endogenous Sensory Shift (O_{t+1})